

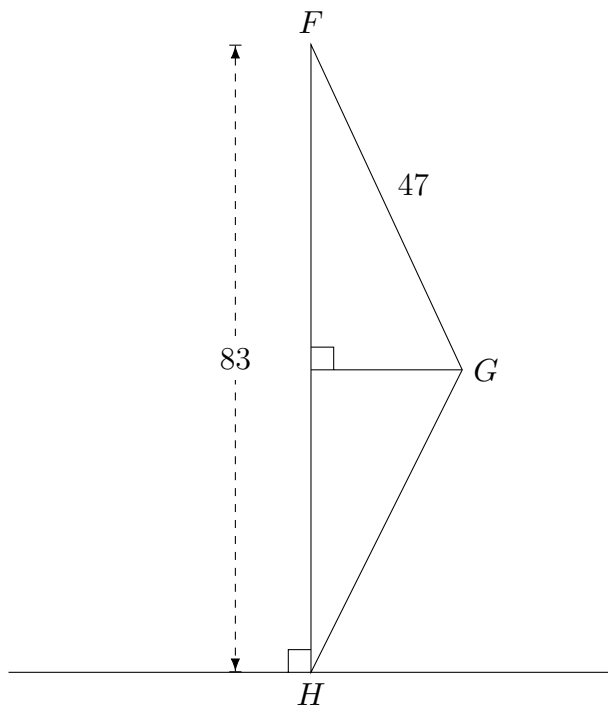
## Standard 2 Trigonometry

### Question Bank

#### Question 1

The triangle  $FGH$  is shown below. The length of  $FG$  is 47 and the length of  $FH$  is 83.

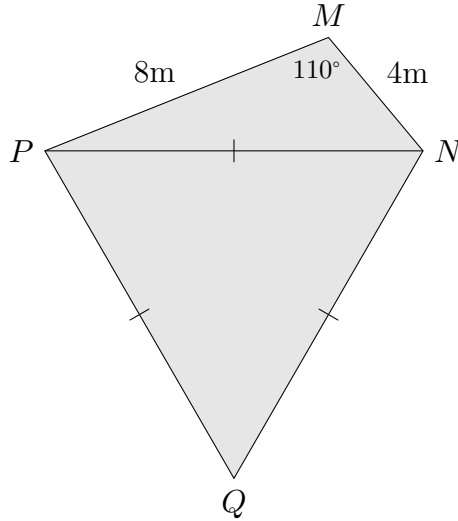
The angle of elevation of  $G$  from  $H$  is  $64^\circ$  and  $\angle FGH$  is obtuse.



Find the length of  $GH$  correct to two decimal places.

### Question 2

A strange window is constructed from two triangles,  $PMN$  and  $PNQ$ . The length of  $PM$  is 8m and the length of  $MN$  is 4m. The  $\angle PMN$  is  $110^\circ$ .



The window is to be coated in a protective film that costs  $\$45/\text{m}^2$ , how much will it cost to coat the window?

## Answers

### Question 1

Since  $\angle FHG$  is complementary to  $64^\circ$ ,

$$\angle FHG = 26^\circ.$$

Find  $\angle FGH$  using sine rule,

$$\begin{aligned}\frac{\sin(\angle FGH)}{83} &= \frac{\sin(26^\circ)}{47} \\ \sin(\angle FGH) &= \frac{\sin(26^\circ)}{47} \times 83 \\ \angle FGH &\approx 50.7^\circ\end{aligned}$$

But since  $\angle FGH$  is obtuse,

$$\angle FGH \approx 180^\circ - 50.7^\circ = 129.3^\circ$$

$$\angle GFH = 180^\circ - 129.3^\circ - 26^\circ = 24.7^\circ$$

$$\begin{aligned}\frac{GH}{\sin(24.7^\circ)} &= \frac{47}{\sin(26^\circ)} \\ GH &= \frac{47}{\sin(26^\circ)} \times \sin(24.7^\circ) \\ &\approx 44.80\end{aligned}$$

### Question 2

The area of  $\triangle PMN$  is

$$A_{\triangle PMN} = \frac{1}{2} \times 4 \times 8 \times \sin(110^\circ) \approx 15.04\text{m}^2$$

The length of  $PN$

$$\begin{aligned}PN^2 &= 4^2 + 8^2 - 2 \times 4 \times 8 \times \cos(110^\circ) \\ PN &= \sqrt{101.889...} \\ &\approx 10.1\text{m}\end{aligned}$$

The area of equilateral  $\triangle PNQ$

$$A_{\triangle PNQ} = \frac{1}{2} \times 10.1^2 \times \sin(60^\circ) \approx 44.17\text{m}^2$$

The cost is

$$(15.04 + 44.17) \times 45 = \$2664.45$$